Rent Control Capitalization into Land Values:

Evidence from New Jersey's New Construction Exemption

Matthew Hockert*

June 30, 2026

Abstract

Rent control ordinances are growing in popularity across the United States, however, they also come with regulatory costs. These costs can often be difficult to quantify due to coverage and changes of the law across different levels of government. This paper exploits New Jersey's New Construction Exemption (NCE), a state law enacted in 1987 that exempts newly constructed buildings with four or more units from local rent control. Because the NCE applies to fourplexes but not triplexes, and given that NJ municipalities set city-specific coverage thresholds, the exemption creates discontinuities in the regulatory burden across building sizes, vintages, and cities. Using a difference-in-difference design on 2024 assessed values, I compare land and improvement values of fourplexes and triplexes, built before and after the NCE, in rent-controlled and never-controlled cities. I find that the NCE premium on land values is approximately 4.7 percent and is significant at all three legislative cutoffs (1987, 1992, 1997). The improvement-value premium is positive but largely mechanical, due to the cost-based assessment practice on buildings. The result survives leave-one-out city robustness, alternative threshold samples, and alternative treated group definitions. These findings indicate that the regulatory cost of rent control capitalizes into land values and less so into improvements.

*Department of Applied Economics, University of Minnesota. Email: hocke069@umn.edu

1 Introduction

Rent control has been experiencing an increase in popularity as states such as California and Oregon both passed statewide rent control ordinances in 2019, while cities of St. Paul, Minnesota (2021); South Portland and Portland, Maine (2022) have enacted their own policies. Both states and cities are enacting these policies as a means to mitigate rising housing costs. While these policies may improve the wellbeing of those that live in those places there are significant costs as well. In theory, rent control constrains the income stream for covered properties and should reduce the market value of those properties relative to uncovered ones. Empirical evidence of this capitalization channel comes primarily from two settings. First, San Francisco, where Diamond et al. (2019) found that rent control reduced rental housing supply by 15 percent and shifted the housing stock toward condominiums and owner-occupied units. Second, Cambridge, Massachusetts, where Autor et al. (2014) found a $1.8 billion gain in assessed value following the removal of rent control in 1995.

Both of these studies utilize periods where rent control was either expanded or eliminated across an entire city. A different source of variation that has not been widely explored comes from exemptions within rent control ordinances (Kholodilin, 2024). Cities may exempt new construction from rent control to preserve development incentives. If theory is correct that rent control imposes significant costs to covered properties then the gap in assessed values between exempt and covered properties should reflect that cost.

Within New Jersey specifically, prior empirical work has reached the opposite conclusion. Gilderbloom and Ye (2007) along with the replication and extension by Ambrosius et al. (2015), compare rent-controlled and non-controlled NJ municipalities using 2000 and 2010 Census data and find no statistically significant effect of rent control on rents, housing quality, property values, or foreclosure rates. The authors interpret this as evidence that New Jersey's ordinances do not meaningfully constrain local housing markets. Their identification does not include the variation in coverage across cities and thus pools differences in their data that would assist in finding a causal effect. The New Construction Exemption is a state law that overrides local ordinances at the four-unit boundary and generates within-city discontinuities in regulatory coverage across building sizes. Non-controlled cities enter my design as a counterfactual for one estimation to eliminate common trends across the groups, but my identification of the capitalization effect comes from variation within municipality-level comparisons.

New Jersey enacted the New Construction Exemption (NCE), in 1987 ("Newly Constructed Multiple Dwellings, Exemption from Rent Control", 1987). The NCE exempts newly constructed multiple dwellings with four or more units from local rent control ordinances. The exemption was initially temporary with five-year sunset, but was extended in 1992, and made permanent in 1997. New Jersey has roughly 112 rent-controlled municipalities, each with their own set of coverage thresholds. The NCE, being a state policy that overrides local ordinances, generates exogenous variation across building sizes, vintages, and cities that I leverage for a causal effect.

Because the NCE exempts buildings with four or more units, triplexes are therefore subject to rent control like the pre-1987 fourplexes but do not receive the exemption like the post-1987 fourplexes, making them a natural within-city control group (Section 2.2). I implement a difference-in-difference (DD) and a triple-difference (DDD) design using 2024 assessed values from the NJ MOD-IV Historical Database (Rutgers Policy Lab, 2024), which reports land and improvement values for every parcel in the state. The two differences are: (1) construction vintage (pre- vs. post-1987), (2) building size (triplex vs. fourplex), (3) and for the DDD compared rent-controlled to never-controlled.

I find that the NCE capitalizes into land values by roughly 4.7 percent. This effect is significant at all three legislative cutoffs and concentrated in land values. The cross-city comparison (triple-difference model) yields a much larger premium (49 percent), but when including census tract fixed effects the gap loses statistical significance. This result highlights that there is sorting between neighborhoods when choosing to develop these specific parcels. The improvement-value premium is positive but largely mechanical, reflecting the cost-based assessment of newer structures rather than a rent-control effect. The result is robust to leave-one-out city tests, alternative threshold samples, and definitions of the treated group.

The paper makes three contributions. First, it provides direct evidence on two of the least-studied outcomes in the rent control literature (Kholodilin, 2024) that the capitalization into property values and the loss to the municipal property tax base. Additionally, the land and improvement decomposition is not available in prior work. Second, my findings show that exemptions from rent control can create large capitalization effects implying that the regulatory cost of coverage is large. Third, it builds on the prior single-city studies by leveraging 43 rent-controlled cities within a single state, each with a different coverage threshold, providing within-state variation in regulatory intensity.

The remainder of the paper proceeds as follows. Section 2 describes the institutional background of rent control in New Jersey and the NCE. Section 3 describes the data. Section 4 presents the empirical strategy. Section 5 reports the main results. Section 6 presents robustness checks. Section 7 discusses the findings, and Section 8 concludes.

2 Institutional Background

2.1 Rent Control in New Jersey

New Jersey permits municipalities to adopt rent control ordinances individually. As of 2024, 112 municipalities have adopted some form of rent control and 82 adopted their ordinances by 1987, prior to the NCE being enacted. Each municipality sets its own coverage threshold, the minimum building size subject to rent control. A city with a threshold of 3, for example, subjects all buildings with three or more units to rent control, while buildings with fewer than three units are exempt. Section 3.3 describes how coverage thresholds are validated and the sample construction.

2.2 The New Construction Multiple Dwelling Law

In 1987, the New Jersey Legislature enacted the New Construction Multiple Dwelling Law (P.L. 1987, c.153), which exempts newly constructed multiple dwellings with four or more units from local rent control ordinances. The exemption lasts for 30 years or the life of the initial mortgage, whichever is longer. To claim the exemption, the building owner must file a notice with the municipality at least 30 days before the certificate of occupancy is issued ("Filing of Owner's Claim of Exemption", 1987). I do not explore whether a property owner claimed the exemption.

The NCE was enacted in three waves. In 1987, the exemption was temporary, with a five-year sunset. In 1992, P.L. 1992, c.206 ("Extension of Newly Constructed Multiple Dwellings Exemption", 1992) extended the exemption with a 15-year sunset. In 1997, P.L. 1997, c.56 ("Amendment Making NJNCMDL Exemption Permanent", 1997), removed the sunset clause and made the exemption permanent. The 4+ unit definition and the 30-year exemption period have not changed since 1987. Each wave of the NCE provide distinct cutoffs that can be tested to understand if each corresponding wave lead to increasing credibility of the exemption.

Table 1 summarizes the three NCE legislative events.

Table 1: NCE Legislative Timeline

Law	1987 P.L. 1987, c.153	1992 P.L. 1992, c.206	1997 P.L. 1997, c.56
Status	Enacted (temporary)	Extended	Permanent
Sunset	5 years	15 years	Removed
Definition	4+ units	unchanged	unchanged
Exemption	30 yr or mortgage	unchanged	unchanged

Notes: P.L. 1997, c.56 is a one-sentence law that deleted the sunset clause. The 4+ unit multiple dwelling definition and 30-year/mortgage exemption period have been unchanged since 1987.

3 Data

3.1 MOD-IV Assessed Values

The data source is the 2024 MOD-IV file from the NJ MOD-IV Historical Database, compiled by the Rutgers Policy Lab (Rutgers Policy Lab, 2024). The database contains parcel-level assessor records for every property in the state. The key fields are property class, number of dwelling units, year constructed, land value, improvement value, and calculated acreage.

A central feature of New Jersey's assessment system is that land and improvement values are reported separately for every parcel. If rent control constrains the income stream from a property I should find land values in exempted buildings to be greater than covered ones.

I restrict the sample in several ways. First, buildings with a recorded construction year of zero or an improvement value of zero are excluded because they most likely represent vacant land or data errors and not residential structures. Second, buildings are classified into four types based on the number of dwelling units: duplexes (2 units), triplexes (3 units), fourplexes (4 units), and large multifamily (5 or more units, or NJ property class 4C). The Class 2 (residential) and Class 4C (commercial apartment) designations in New Jersey's assessment system create a natural boundary at four units where triplexes and fourplexes are nearly all Class 2, while five-plexes and above are nearly all Class 4C (Table 8). Third, the outcome variables are log land value, log improvement value, and log total assessed value to interpret the coefficients as percentage changes. Parcel size is measured as log acreage.

3.2 How New Jersey Assessors Value Land and Improvements

In New Jersey assessors determine the value of a parcel in three ways: (1) the income approach which is where the assessor estimates the property's worth through its ability to produce income. This is largely done by dividing the annual net income by the appropriate capitalization rate. However, obtaining all the necessary data to do this estimation is difficult and assessors may fall back on other methods; (2) the sales approach, instead of estimating future cash flows like the income approach, the assessor instead uses sale values of similar properties nearby and then adjusts for other qualities of the property; (3) the cost approach which requires the assessor to estimate replacement cost of the new building from standardized unit-cost schedules and physical building characteristics, then subtracts depreciation. For two-to-four unit buildings the structure is valued by the cost approach where replacement costs are calculated using the number of units and floor area per story minus depreciation. The land is valued separately, but by one of several methods. If vacant-land sales exist, a land value map assigns unit values (e.g. per front foot, square foot, or acre) by location and where they do not assessors use the allocation method (land as a share of total value inferred from improved sales), the abstraction method (land as the improved sale price net of depreciated building cost), or the land-residual method (the capitalized income attributable to land) (New Jersey Division of Taxation, 2021).

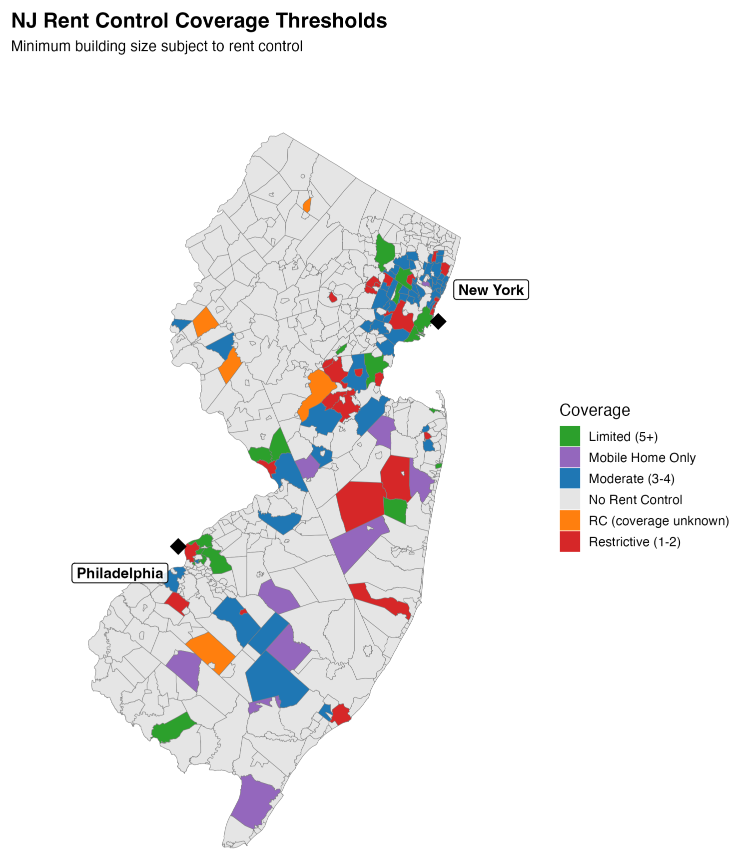
Because the structure is valued using methods that do not account for future cash flows the effects on improvements should be largely invariant to a rent-control restriction. However, the land component is different: the allocation, abstraction, and land-residual methods each back land value out of the property's market sale or income. This means that if the exemption raises a building's market value that premium should mechanically be allocated into the assessed land value but under a pure cost-plus-vacant-land-map valuation it would not be. To better control for these varying appraisal methods and the possible relationship within neighborhoods I estimate the models using census tracts as a fixed effect in Section 5.

3.3 Coverage Thresholds

Of the 112 municipalities that have adopted rent control, 82 did so by 1987 (before the NCE). Coverage thresholds were validated through Open Public Records Act (OPRA) requests sent to all 82 pre-1987 cities. Twenty-one cities returned historical ordinance text, which I use as the primary source. For the remaining cities, I extract the coverage threshold from the oldest available modern ordinance text. Of the 21 cities with OPRA-verified historical ordinances, only 4 (19 percent) had a coverage threshold that differed from the modern ordinance text, suggesting that thresholds are generally stable over time.

Of the 82 pre-1987 cities, 8 have mobile-home-only ordinances or lack sufficient documentation for a residential coverage threshold, leaving 74 cities with validated thresholds. Three additional cities are excluded because their ordinances were repealed or never enforced, leaving 71 in the analysis sample. The threshold distribution across the 74 validated cities is: threshold 1 (20 cities), threshold 2 (5 cities), threshold 3 (21 cities), threshold 4 (18 cities), and threshold 5 or above (10 cities). At thresholds 1 through 3, both triplexes and fourplexes are covered by rent control. At threshold 4, fourplexes are covered but triplexes are not. At threshold 5 or above, neither triplexes nor fourplexes are covered. Figure 1 maps these across the state.

Figure 1: Rent Control Coverage Thresholds Across New Jersey Municipalities



3.4 Summary Statistics

The sample is constructed from the 2024 MOD-IV cross-section as follows. I restrict to rent-controlled cities at thresholds 1 through 3 and all never-controlled cities. Within these cities, I keep only triplexes and fourplexes, and drop observations with missing or infinite log acreage. The resulting sample contains 30,384 buildings across 419 cities (43 rent-controlled and 376 never-controlled).

Figure 2 maps the cities by their adoption being before or after 1987. A city is classified as rent-controlled if it adopted its ordinance by 1987 and has a validated coverage threshold.

A city is classified as never-controlled if it has never adopted a rent control ordinance. All other cities are excluded.

Figure 2: DDD Analysis Sample: Pre-1987 RC Cities and Never-RC Baseline

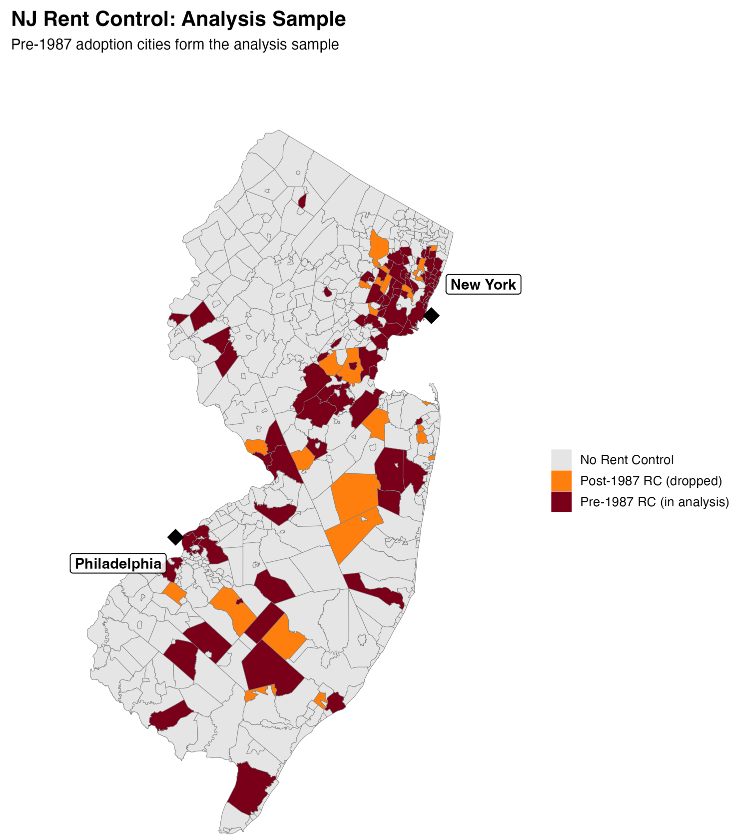


Table 2 reports the cell sizes for the sample. The treated cell contains 130 post-1987 fourplexes across 15 rent-controlled cities (see Appendix Table 11 for a breakdown by threshold). The fourplex cells in the post-1987 period is small in both rent-controlled and never rent-controlled cities reflecting a common trend of a lower density building trend.

Table 2: Observation by Cell (Threshold 1–3 RC + Never-RC)

		RC (thresh 1–3)	Never RC
Triplexes	Pre-1987	13,431 (42)	7,176 (356)
	Post-1987	2,452 (25)	251 (102)
Fourplexes	Pre-1987	4,329 (42)	3,616 (315)
	Post-1987	130 (15)	163 (80)

Notes: Post-1987 fourplexes in RC cities at thresholds 1–3. Parentheses show the number of cities contributing to each cell.

Table 3 reports summary statistics by cell. Post-1987 fourplexes in rent-controlled cities have mean land values of $179,505 on lots averaging 0.09 acres, built around 2001. Their never-RC counterparts have substantially higher land values ($584,486) but also much larger lots (77 acres). The differences in lot size and values is not a cause for concern in the with-RC DD models as much of that development happens within neighborhood on similar infill lots. This is observed in the within the post-1987 cohort, triplexes and fourplexes are compositionally similar in both panels with small-lots and built around 2001–2004. However, the DDD controls for lot size and vintage to prevent the level differences from biasing the estimate.

Table 3: Summary Statistics

			Assessed Value ($)			
Group	Cell	N	Mean Land	Mean Imp	Mean Acres	Mean Yr
RC (threshold 1–3, 43 cities):						
	Pre-87 triplex	13,431	94,991	221,544	10.69	1913
	Post-87 triplex	2,452	75,268	322,968	1.82	2004
	Pre-87 fourplex	4,329	208,484	357,595	41.23	1917
	Post-87 fourplex	130	179,505	608,729	0.09	2001
Never RC (356 cities):						
	Pre-87 triplex	7,176	207,171	260,026	1.194	1916
	Post-87 triplex	251	568,331	936,092	1.975	2003
	Pre-87 fourplex	3,616	203,255	303,971	6.690	1907
	Post-87 fourplex	163	584,486	1,251,205	77.104	2001

Notes: 2024 MOD-IV assessed values. Cell sizes from the threshold 1–3 pooled sample (43 RC cities, 356 never-RC cities).

Table 4 shows the post-1987 multifamily construction mix by coverage threshold. At thresholds 3 and above duplexes dominate (84 to 94 percent) likely because they fall below the coverage threshold. At thresholds 1 and 2, where duplexes are also covered, triplexes hold approximately 30 percent of new construction. Fourplexes make up only 1 to 2 percent at all thresholds. Never-controlled cities show a similar mix to high-threshold cities (87 percent duplexes, 3 percent fourplexes). This pattern is reassuring for identification as it highlights a lack of a selection issue in that the NCE did not induce a wave of fourplex building, and the post-1987 fourplexes I compare are not a sample selected into the policy.

Table 4: Post-1987 Multifamily Construction Mix by Coverage Threshold

Thresh	Cities	Dup	Tri	Four	5–10	11+	4C Apt	N
1	18	63.9%	32.2%	1.4%	0.2%	0.5%	1.8%	6,235
2	4	62.6%	28.6%	1.0%	0.7%	1.0%	6.1%	297
3	20	83.9%	10.7%	1.0%	0.7%	1.2%	2.5%	3,371
4	18	85.1%	8.8%	1.6%	0.6%	1.2%	2.8%	2,150
5	5	93.8%	3.3%	0.4%	0.1%	—	2.4%	4,760
Never RC		86.7%	4.3%	2.8%	1.0%	0.8%	4.5%	5,849

Notes: Share of new 2+ unit buildings constructed after 1987, from 2024 MOD-IV stock. 4C Apt = Class 4C apartment buildings (non-condo) with unit counts not recorded in MOD-IV. Excludes condominiums.

4 Empirical Strategy

4.1 Difference-in-Differences Specifications

If rent control coverage imposes a cost on covered properties, the NCE should raise the value of exempt buildings relative to covered ones. Post-1987 fourplexes in rent-controlled cities receive the exemption while triplexes in the same cities do not. This paper runs two separate differences-in-differences (DD) models. The first model compares the fourplex-triplex value gap by construction vintages (pre- vs. post-1987). In these threshold 1-3 cities, triplexes are covered before and after 1987 while fourplexes built before 1987 are covered, but those built after are exempted. The second model is a triple-difference (DDD) where I extend the framework by comparing between city types (rent-controlled vs. never-controlled) with the goal of isolating the RC-specific NCE premium and netting out any common vintage-by-size trends that may have occurred throughout New Jersey.

The DDD is an important addition because the fourplex-triplex value gap may shift at 1987 for reasons unrelated to rent control. The DD on never-controlled cities captures any such common trends. However, the never rent-controlled cities are different from the rent controlled cities in that they tend to be more suburban and rural than the more urban rent controlled cities. Therefore, I include both the DD and DDD specifications below:

Equation 1 below is the DD specification:

$$\begin{aligned} \ln(\text{Value}_i) = \alpha_c &+ \beta_1 \cdot \text{Post}_i + \beta_2 \cdot \text{Fourplex}_i \\ &+ \beta_3 \cdot (\text{Post}_i \times \text{Fourplex}_i) \\ &+ \phi \cdot \ln(\text{acres}_i) + \gamma_{\text{pre}} \cdot \tilde{Y}_i^- + \gamma_{\text{post}} \cdot \tilde{Y}_i^+ + \varepsilon_i \end{aligned} \tag{1}$$

Equation 2 below is the DDD specification:

$$\begin{aligned} \ln(\text{Value}_i) = \alpha_c &+ \beta_1 \cdot \text{Post}_i + \beta_2 \cdot \text{Fourplex}_i + \beta_3 \cdot \text{RC}_i \\ &+ \beta_4 \cdot (\text{Post}_i \times \text{Fourplex}_i) + \beta_5 \cdot (\text{Post}_i \times \text{RC}_i) + \beta_6 \cdot (\text{Fourplex}_i \times \text{RC}_i) \\ &+ \delta^{DDD} \cdot (\text{Post}_i \times \text{Fourplex}_i \times \text{RC}_i) \\ &+ \phi \cdot \ln(\text{acres}_i) + \gamma_{\text{pre}} \cdot \tilde{Y}_i^- + \gamma_{\text{post}} \cdot \tilde{Y}_i^+ + \varepsilon_i \end{aligned} \tag{2}$$

The outcome is log assessed value (total, improvement, or land) for parcel i. α_c denotes census tract fixed effects. Post_i equals one if the building was constructed at or after an NCE cutoff (1987, 1992, or 1997), Fourplex_i equals one if the building has four units, and RC_i equals one if the city adopted rent control by 1987 with a validated coverage threshold of three or below. $\tilde{Y}_i^-$ and $\tilde{Y}_i^+$ are piecewise linear vintage slopes centered at the cutoff, allowing separate pre- and post-cutoff trends in construction year which is supported by the visuals of the binned estimates found in the Results and Appendix B sections. In the DiD model, β_3 represents the causal estimate of the value premium between the treated and control groups. In the DDD specification, the coefficient β_4 is the DD on never-RC cities which is the change in the fourplex–triplex value gap at the cutoff in cities without rent control. δ^{DDD} is the triple difference coefficient measuring how much more the fourplex–triplex gap shifts at the cutoff in RC cities relative to never-RC cities. The sum $\beta_4 + \delta^{DDD}$ returns the DD on RC cities which is the change in the fourplex–triplex gap at the cutoff within rent-controlled cities and is estimated in Equation 1. I provide the estimates for both models in the main results table 5. Standard errors are clustered at the city level.

5 Results

Table 5 below provides the estimates from equations 1 and 2 in the methods section. I find varying effects by cutoff year and outcome variable but also on whether the identification is modeled as within-RC or using the never-RC cities as an additional control group to the RC cities. Table 5 reports only the headline DD and triple-difference coefficients; the full set of coefficients for every model is reported under census-tract fixed effects in Appendix C and under city fixed effects in Appendix D.

Table 5: Results: Threshold 1–3 Cities (Census-Tract FE)

Cutoff	Outcome	DD (within-RC)	DD (never-RC)	$\hat{\delta}^{DDD}$
1987	log(Total)	0.090***	0.063	0.040
		(0.032)	(0.065)	(0.072)
	log(Imp)	0.138**	0.151*	0.005
		(0.059)	(0.089)	(0.106)
	log(Land)	0.046**	−0.026	0.086
		(0.017)	(0.056)	(0.060)
1992	log(Total)	0.089**	0.067	0.034
		(0.033)	(0.081)	(0.087)
	log(Imp)	0.151**	0.142	0.021
		(0.063)	(0.116)	(0.131)
	log(Land)	0.037**	−0.012	0.063
		(0.015)	(0.066)	(0.069)
1997	log(Total)	0.112***	0.095	0.026
		(0.034)	(0.087)	(0.094)
	log(Imp)	0.189**	0.168	0.028
		(0.073)	(0.125)	(0.143)
	log(Land)	0.057***	0.015	0.057
		(0.015)	(0.071)	(0.074)
Diagnostics				
N		19,556	29,288	
RC cities		43	43	
Never-RC cities		—	365	
Fixed Effects		Census tract	Census tract	
Census tracts		341	949	
Clustering		City	City	
Within R^2		0.249 / 0.223 / 0.092	0.240 / 0.190 / 0.145	

Notes: Census-tract fixed effects. DD (within-RC) is from Equation 1 and estimated on RC cities only; DD (never-RC) and $\hat{\delta}^{DDD}$ are from Equation 2 pooling RC and never-RC cities. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

The within-RC DiD (Equation 1) compares the fourplex–triplex value gap across construction vintages within rent-controlled cities. The 1987 cutoff is the original law exempting newly constructed four-unit dwellings from rent control. At this cutoff, the total assessed value (land plus improvements) of exempt fourplexes is 9.4 percent higher than that of covered triplexes ($e^{0.090} - 1$). This total effect is driven mainly by improvement value, where exempt fourplexes are 14.8 percent higher. As discussed in Section 3.2, that improvement gap is largely mechanical rather than a rent-control effect. This is due to the post-1987 structures being newer and given the cost approach of valuations these properties have depreciated less. The relevant test for capitalization is therefore the land component, to which I find the gap between exempt fourplexes and covered triplexes to be 4.7 percent higher than pre-1987 properties. This increase is largely attributed to rent control due to assessment methodologies. As mentioned above, the land assessment practice uses three primary ways to value the land portion of a property: allocation, abstraction, and land residual. Each of these methods backs land out of the property's total market value, leaving land as the residual that absorbs changes in market value. Because the structure is held at depreciated replacement cost under the cost assessment approach, it cannot reflect the income gain from the exemption and therefore the higher market value an exempt building commands is therefore attributed entirely in its land.

The never-RC and DDD columns come from the triple-difference in Equation 2. In the never-RC column, which reflects the fourplex–triplex gap in cities that never adopted rent control, only the estimation on improvement value is significant (0.151, $p < 0.10$), while total and land values are null. The triple difference nets out this never-RC trend from the within-RC estimate. The DDD on total and improvement value are null while the land effect is positive (0.086) but not statistically significant. The improvement premium is essentially the same in rent-controlled and never-controlled cities at 0.005, which is additional evidence that it is a mechanical effect rather than pure capitalization. The never-controlled estimate is only marginally significant at 1987, but the difference of the two columns is what identifies the improvement effect as non-capitalization. It is also important to highlight that the never rent-controlled cities are quite different from those that do have rent control. Never-controlled cities tend to be more suburban and rural, larger lots, and lower density (Table 3). This means that they tend to be an imperfect counterfactual for the small-lot urban infill that rent-controlled cities produce. Census-tract fixed effects and the log(acres) control absorb part of this gap but most likely not all of it. Because of these structural difference, I treat the within-RC comparison as the main results and the DDD as a check on common trends. The coefficients identified in the never-RC and the DDD columns in 1987 hold over subsequent cutoff years.

The following cutoff years, show that total and improvement values stay positive and statistically significant. While the total value gap over cutoffs varies $(0.084, 0.079, 0.097)$ the gap on improvement values continued to increase $(0.130, 0.147, 0.181)$. However, the statistical significant of the land value estimate is not significant in 1992, when the law was only extended for another 5 years. This may reflect the pricing in of the one time short extension of the NCE. However, the 1997 cutoff finds a similar size effect on land values as 1987 but less statistically significant (0.043, $p < 0.10$).

These results, depending on the outcome variable, highlight three important lessons. The first is that improvement value gap being large, positive, and continued to climb monotonically over each subsequent cutoff period shows that assessor methodology is being captured in improvements. The second is that land values vary by size and statistical significance over cutoff years illustrating the implications of the role that rent-control prices in future expected returns on exempted properties. Lastly, census tract fixed effects remove much of the variation within neighborhoods of cities regardless of rent control thus producing smaller estimates and a lack of statistical significance in the triple difference (DDD) estimator. In Appendix D I re-estimate the identical models shown in this section only using city fixed effects. I find a different story, where the within-RC estimates tend to be larger, positive, and variation in statistical significance by outcome and cutoff year. I also find that the DDD estimate on land to be statistically significant at every cutoff and the effects to range from 49.3 percent to 52.7 percent depending on cutoff year. The gap on total and improvement value is not statistically significant. The results presented in Table 5 are sensitive to adjustments and the following robustness section presents numerous tests to highlight the sensitivity of the main results.

6 Robustness

The main result relies on 43 rent-controlled cities at thresholds 1 through 3. This section tests the sensitivity of the within-RC DD across multiple methodological choices I made. In the first section, I vary the threshold sample to test whether the result holds within threshold groups. Next, I run a leave-one-out (LOO) test where I drop each rent-controlled city one at a time to verify that no single city drives the result. I proceed to then expand the units above fourplexes to test whether the NCE premium extends to larger buildings. Lastly, I re-estimate the within-RC DD with a series of pre-NCE placebo cutoffs.

6.1 Threshold Group Robustness

Table 6 reports the within-RC DD by outcome across threshold subgroups. Total and improvement values are significant across nearly all threshold groups and cutoffs. The within-RC land effect is significant at every cutoff in the pooled threshold equal to 1–3 (0.046) and threshold equal to 3 (0.064) samples. The effect is not statistically significant at threshold equal to 1 (0.015) and threshold equal to 1–2 (0.028). These sets also have the lowest number of observations. The triple difference was not significant across cutoffs nor outcome.

Table 6: Within-RC DD Across Threshold Groups, by Outcome (Census-Tract FE)

Sample	Outcome	1987	1992	1997
Thresh 1–3 pooled	log(Total)	0.090*** (0.032)	0.089** (0.033)	0.112*** (0.034)
(43 cities)	log(Imp)	0.138** (0.059)	0.151** (0.063)	0.189** (0.073)
	log(Land)	0.046** (0.017)	0.037** (0.015)	0.057*** (0.015)
Thresh=3 only	log(Total)	0.170** (0.067)	0.170** (0.076)	0.171** (0.077)
(21 cities)	log(Imp)	0.229* (0.120)	0.236* (0.129)	0.237* (0.130)
	log(Land)	0.064** (0.025)	0.072** (0.033)	0.072** (0.033)
Thresh=1 only	log(Total)	0.078** (0.033)	0.087** (0.034)	0.122*** (0.033)
(17 cities)	log(Imp)	0.128** (0.060)	0.150** (0.066)	0.207** (0.087)
	log(Land)	0.015 (0.023)	0.010 (0.025)	0.036 (0.033)
Thresh 1–2	log(Total)	0.068* (0.033)	0.074** (0.034)	0.103*** (0.033)
(22 cities)	log(Imp)	0.103 (0.060)	0.131** (0.061)	0.179** (0.077)
	log(Land)	0.028 (0.021)	0.015 (0.020)	0.038 (0.027)
Diagnostics				
Fixed Effects			Census tract	
Clustering			City	

Notes: Within-RC DD (Post × Fourplex) on RC cities only at each coverage threshold, census-tract FE, tract-assigned subset; coefficient with clustered SE in parentheses. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

6.2 Leave-One-Out City Robustness

I next run a leave one out test where I drop each of the 43 rent-controlled cities one at a time, re-estimate the within-RC DD on the remaining 42, and report how many of the 43 leave-one-out estimates clear the 10 percent level along with the cluster jackknife t (Table 7). Total, improvement, and land values all clear 10 percent in every one of the 43 drops at each cutoff, so no single city drives the result. The jackknife t-statistics are weaker: below 1.5 for total and improvement and for land at 1987 and 1992, but reaching 2.70 for land at 1997. The within-RC land premium is significant under the conventional clustered standard error and, at the 1997 cutoff, also under the more conservative jackknife.

Table 7: Leave-One-Out Within-RC DD, by Outcome (Census-Tract FE)

Outcome		1987	1992	1997
log(Total)	Baseline	0.090	0.089	0.112
	Range	[0.069, 0.166]	[0.069, 0.180]	[0.092, 0.208]
	Sig at 10%	43/43	43/43	43/43
	Jackknife t	1.12	0.94	1.13
log(Imp)	Baseline	0.138	0.151	0.189
	Range	[0.101, 0.218]	[0.111, 0.264]	[0.144, 0.311]
	Sig at 10%	43/43	43/43	43/43
	Jackknife t	1.43	1.20	1.36
log(Land)	Baseline	0.046	0.037	0.057
	Range	[0.037, 0.080]	[0.031, 0.067]	[0.051, 0.071]
	Sig at 10%	43/43	43/43	43/43
	Jackknife t	1.24	1.12	2.70

Notes: Within-RC DD (Post × Fourplex) on RC cities only, census-tract FE; each of the 43 RC cities dropped once. "Sig at 10%" counts how many of the 43 leave-one-out estimates clear the 10% level; the last row of each block is the jackknife t. Tract-assigned subset; clustered at city.

6.3 Treated Group Definition

The main specification defines the treated group as fourplexes only. This choice is due to the structure of New Jersey's property classification system where duplexes to fourplexes are classified together (Class 2) and buildings with five or more units are classified as Class 4C (commercial apartment). These two classes are different property types and have different assessment practices and market dynamics, such as being able to account for future cash flows in their valuations (New Jersey Division of Taxation, 2021). Table 8 below shows the number of units in the 43 sample of rent controlled cities prior to the 1987 cutoff and after. The table highlights a general slow down in development across both classes following the 1987 NCE mandate.

Table 8: NJ Building Stock by Unit Size and Property Class (2024 MOD-IV)

	Class 2 (Residential)		Class 4C (Comm. Apt)	
Units	RC pre-87	RC post-87	RC pre-87	RC post-87
3	20,200	2,675	43	5
4	6,628	163	155	3
5	26	1	302	9
6	20	1	484	11
7	0	1	129	4
8	7	1	175	6
9	0	0	60	4
10	2	0	42	5

Notes: Pre-1987 RC cities with validated coverage thresholds (all thresholds), excluding condominiums.

Table 9 below shows the coefficient by outcome and cutoff year when I expand the treated group beyond fourplexes. I first test the fourplex plus five unit buildings as a treated group and then expand to all buildings with four or more units. The within-RC DD grows with building size, with the 1987 land coefficient rising from 0.046 for fourplexes to 0.35 and 0.38 as larger buildings are developed. All three definitions are significant at every cutoff. This gradient is largely mechanical, as larger exempt buildings contain more units and therefore more foregone rent that would depreciate the land value. Fourplexes remain the preferred treated group because fourplexes and triplexes use the same assessment practices and are directly on either side of the NCE mandated unit cutoff.

Table 9: Within-RC DD: Alternative Treated Group Definitions, by Outcome (Census-Tract FE)

Treated group	Outcome	1987	1992	1997
Fourplexes only (Class 2)	log(Total)	0.090***	0.089**	0.112***
	log(Imp)	0.138**	0.151**	0.189**
	log(Land)	0.046**	0.037**	0.057***
4–5 units (Class 2 + 4C)	log(Total)	0.349***	0.380***	0.417***
	log(Imp)	0.374***	0.409***	0.455***
	log(Land)	0.349**	0.382*	0.420**
4+ units (all, incl. 4C)	log(Total)	0.366***	0.396***	0.440***
	log(Imp)	0.411***	0.445***	0.497***
	log(Land)	0.384***	0.423***	0.461***
Diagnostics				
Fixed Effects			Census tract	
Clustering			City	

Notes: Within-RC DD (Post × NCE-eligible) on RC cities only across the three NCE cutoffs, threshold 1–3 sample, control group triplexes; census-tract FE, tract-assigned subset. Tract-assigned N: 29,288 (fourplexes), 31,606 (4–5 units), 34,800 (4+ units). Clustered at city. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

6.4 Pre-NCE Placebo Cutoffs

My last robustness test, I replace the true 1987 cutoff with a series of pre-NCE placebo cutoffs and re-estimate the within-RC land DD on the pre-1987 building stock, where no exemption existed. A genuine NCE effect should not appear at these dates. The within-RC land DD is positive and significant at the two earliest cutoffs (0.098 at 1965, 0.127 at 1970) but null at 1975, 1980, and 1985 (Table 10), indicating a pre-existing fourplex–triplex land divergence in the oldest segment of the rent-controlled building stock that fades at later cutoffs.

Table 10: Within-RC DD at Pre-NCE Placebo Cutoffs, by Outcome (Census-Tract FE)

Placebo cutoff	log(Total)	log(Imp)	log(Land)
1965	0.031 (0.026)	−0.005 (0.037)	0.098** (0.038)
1970	0.039 (0.044)	0.006 (0.048)	0.127* (0.065)
1975	0.015 (0.072)	−0.022 (0.075)	0.110 (0.079)
1980	0.085 (0.094)	0.058 (0.092)	0.173 (0.119)
1985	−0.098 (0.080)	−0.143* (0.080)	0.034 (0.077)
1987 (true)	0.090*** (0.032)	0.138** (0.059)	0.046** (0.017)

Notes: Within-RC DD (Post × Fourplex) on RC cities only, pre-1987 building stock, census-tract FE; coefficient with clustered SE. The 1987 (true) row uses the full RC lead sample. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

7 Discussion

I find that New Jersey's New Construction Exemption capitalizes into land values producing a premium of roughly 4.7 percent. The improvement-value premium is more robust but largely mechanical, due New Jersey assessment methodology using the cost approach and newer exempt buildings carry higher improvement values regardless of whether they are rent controlled. The capitalization of the exemption therefore concentrates in land, where the assessor's allocation, abstraction, and land-residual methods soak up much of the market value.

I use the within-RC census-tract-fixed-effects land coefficient to estimate the fiscal cost of rent control through property taxes inspired by the approach of Autor et al. (2014), who estimated the aggregate value gains from decontrol in Cambridge, Massachusetts. Because New Jersey assesses land and improvement values separately, and because this within-neighborhood specification identifies the capitalization effect on land, the missing tax base attributable to rent control can be calculated directly.

In order to do this calculation, I take the 19,643 covered buildings from the regression sample (triplexes and pre-1987 fourplexes in the 43 rent-controlled cities at thresholds 1 through 3) and sum their assessed land values, which total \$2.26 billion in 2024. Using the preferred within-RC land coefficient (0.0455), the implied counterfactual land value without rent control is $e^{0.0455} - 1 = 4.7$ percent higher than what is currently assessed. The missing land assessed value is therefore \$2.26 billion $\times$ 0.0466 = \$105.2 million.

To find the lost tax revenue by city, I multiply each municipality's missing land value by its own 2024 general tax rate (New Jersey Division of Taxation, 2024). New Jersey's general tax rate applies directly to assessed values. Rates vary substantially across the 43 cities (assessed-value-weighted average: 2.54 per \$100). Summing across municipalities yields an implied lost annual property tax revenue of approximately \$2.67 million from land alone, or roughly \$136 per covered building per year. The five largest contributors are Newark (\$638,000), Hoboken (\$453,000), Union City (\$370,000), Elizabeth (\$270,000), and West New York (\$197,000).

This estimate is a lower bound on the total lost revenue for several reasons. First, it reflects only the land component of assessed value. The within-RC effect on total assessed value is positive and significant (0.090), which is much larger than the land effect alone, thus suggesting the full fiscal cost would be much larger. Second, it covers only triplexes and fourplexes, excluding larger multifamily buildings that are also covered by rent control. Third, it applies the capitalization coefficient uniformly across cities, whereas the effect certainly varies across municipalities due to coverage intensity differences. The estimate does not account for other effects of rent control that may have implications on the broader tax base or on municipal expenditures.

These results are comparable to the findings in Diamond et al. (2019) and Autor et al. (2014). Diamond et al. (2019) found that rent control in San Francisco reduced rental housing supply by 15 percent, with landlords converting covered properties to condominiums or other uses. Autor et al. (2014) estimated \$1.8 billion in assessed value gains following the removal of rent control in Cambridge, Massachusetts. This paper complements these findings by showing that even a partial exemption, which removes rent control from new construction above a size threshold, generates large capitalization effects on land values. The magnitude of the fiscal estimate (\$2.67 million per year) is considerably smaller than the \$1.8 billion found by Autor et al. (2014). These value differences are expected given that the present estimate reflects a partial exemption applied to a subset of building sizes across a mix of smaller and larger New Jersey cities and not a complete decontrol in a single high-value market like Cambridge, Massachusetts.

These results also are in tension with prior New Jersey rent control literature. Gilderbloom and Ye (2007) and Ambrosius et al. (2015) compare rent-controlled and non-controlled New Jersey municipalities at the city level and report null effects on rents, housing quality, and property values and thus conclude that the ordinances do not meaningfully constrain local housing markets. However, there are several design features that prevent a meaningful test of capitalization. First, their property value outcome is the percentage change in self-reported median home value between the 2000 and 2010 Censuses. The main identification issues with census data in this instance is that it uses self-reported coverage of owner-occupied housing, which is not subject to rent control and is the set of housing through which rent control should capitalize. Second, the majority of New Jersey rent control ordinances were already in place by 2000. Any effect of rent control due to a change between 2000 and 2010 would have capitalized prior and the authors are most likely estimating the effect of some other confounder. Third, their treatment is either a city-level indicator or a ten-component ordinal index in which the coverage threshold receives one-tenth weight. Pooling coverage with nine other features attenuates the estimate. My within-RC estimates instead show a premium concentrated in land (roughly 4.7 percent) which cannot be detected using their methodology.

There are several limitations to this paper. First, if rent-controlled cities zone against fourplexes but not triplexes, the design could capture zoning restrictions rather than rent control capitalization. City and census-tract fixed effects help address this concern, but they do not fully rule it out. Verification would require historical zoning maps from the 1987 era, which are not currently available in digitized form. Second, New Jersey municipalities revalue property on different schedules. City fixed effects absorb cross-city differences in assessment practices, but within-city variation in assessment timing could introduce noise. Third, the 2022 *Willow Ridge* ruling ("Willow Ridge Apartments, LLC v. Union City Rent Stabilization Board", 2022) established that owners who fail to file for the NCE forfeit the exemption, introducing compliance risk that may depress post-1987 fourplex values and attenuate the effect. Filing status is not recorded in MOD-IV and would require additional OPRA requests to verify at the parcel level. Fourth, the design assumes no spillovers between exempt and covered buildings. Autor et al. (2014) found that decontrolled buildings in Cambridge raised the values of nearby properties that were never under rent control, implying positive spillovers from exemption. If NCE-exempt fourplexes similarly raise the values of nearby covered triplexes, the fourplex-triplex gap underestimates the true NCE premium.

A related concern is selection of post-1987 fourplexes into the NCE. Because the exemption applies only to buildings with four or more units, developers may select into the fourplex size class precisely because of the exemption, in which case the design captures both the value of the exemption and the value of marginal projects shifted into compliance with the four-unit threshold. The construction mix in Table 4 addresses the extensive margin: fourplexes make up only 1 to 2 percent of post-1987 construction in rent-controlled cities, no higher than in never-controlled cities, indicating that the scarcity of fourplexes is a structural feature rather than post-1987 regulatory avoidance. The intensive margin, whether the fourplexes that are built post-1987 in RC cities differ in unobserved quality from those built in never-RC cities, is not directly testable in the current design. A conditional logit of developer building-size choice (two, three, four, or five-plus units) conditional on parcel attributes, estimated separately pre- and post-1987 in RC and never-RC cities, would provide an explicit estimate of how the relative attractiveness of fourplexes shifted at the NCE cutoff which is left for possible future work.

8 Conclusion

This paper estimates the capitalization of rent control into property values using New Jersey's New Construction Exemption. The law exempts newly built fourplexes from local rent control ordinances while leaving triplexes covered. Using 2024 assessed values, I find that, while holding neighborhood fixed, the NCE raises the land value of exempt fourplexes by roughly 4.7 percent relative to covered triplexes in rent-controlled cities. The effect of rent control is concentrated in land rather than improvements and is stable across the three NCE legislative cohorts: 1987, 1992, and 1997. A cross-city comparison yields a much larger premium (roughly 49 percent), but the within-neighborhood estimate shows that most of that gap is due to sorting across neighborhoods. The result survives leave-one-out city robustness, alternative threshold samples, and alternative treated group definitions.

The findings have real implications for the design of rent control policies. The within-neighborhood land value premium shows that rent control coverage imposes a significant cost on covered properties and increased tax burdens on non-covered ones, even in a setting where coverage thresholds are higher than in other cities. The concentration of the effect in land values and not in improvement values, suggesting that rent control costs move through the income channel. I cannot confirm that the effect also leads to physical deterioration of the housing stock but is a finding in other rent control studies. Lastly, the consistent premium across NCE waves indicates that property markets price regulatory exemptions quickly, even when the exemption's permanence is uncertain. As cities and states across the United States impose rent control, the burdens of the policy outlined in this paper should be considered.

References

Ambrosius, J. D., Gilderbloom, J. I., Steele, W. J., Meares, W. L., & Keating, D. (2015). Forty years of rent control: Reexamining New Jersey's moderate local policies after the great recession. *Cities*, *49*, 121–133. https://doi.org/10.1016/j.cities.2015.08.001

Amendment Making NJNCMDL Exemption Permanent. (1997).

Autor, D. H., Palmer, C. J., & Pathak, P. A. (2014). Housing market spillovers: Evidence from the end of rent control in Cambridge, Massachusetts. *Journal of Political Economy*, *122*(3), 661–717. https://doi.org/10.1086/675536

Diamond, R., McQuade, T., & Qian, F. (2019). The effects of rent control expansion on tenants, landlords, and inequality: Evidence from San Francisco. *American Economic Review*, *109*(9), 3365–3394. https://doi.org/10.1257/aer.20181289

Extension of Newly Constructed Multiple Dwellings Exemption. (1992).

Filing of Owner's Claim of Exemption. (1987).

Gilderbloom, J. I., & Ye, L. (2007). Thirty Years of Rent Control: A Survey of New Jersey Cities. *Journal of Urban Affairs*, *29*(2), 207–220. https://doi.org/10.1111/j.1467-9906.2007.00334.x

Kholodilin, K. A. (2024). Rent control effects through the lens of empirical research: An almost complete review of the literature. *Journal of Housing Economics*, *63*, 101983. https://doi.org/10.1016/j.jhe.2024.101983

New Jersey Division of Taxation. (2021). *Real property appraisal manual for New Jersey assessors* (Third, tech. rep.). New Jersey Department of the Treasury, Division of Taxation. Trenton, NJ. https://www.nj.gov/treasury/taxation/pdf/lpt/realpropertyappraisal.pdf

New Jersey Division of Taxation. (2024). *2024 general tax rates* (tech. rep.). New Jersey Department of the Treasury.

Newly Constructed Multiple Dwellings, Exemption from Rent Control. (1987).

Rutgers Policy Lab. (2024). *Introducing the New Jersey MOD-IV historical database* (tech. rep.). Rutgers University.

Willow Ridge Apartments, LLC v. Union City Rent Stabilization Board. (2022).

A Fourplex Detail by Coverage Threshold

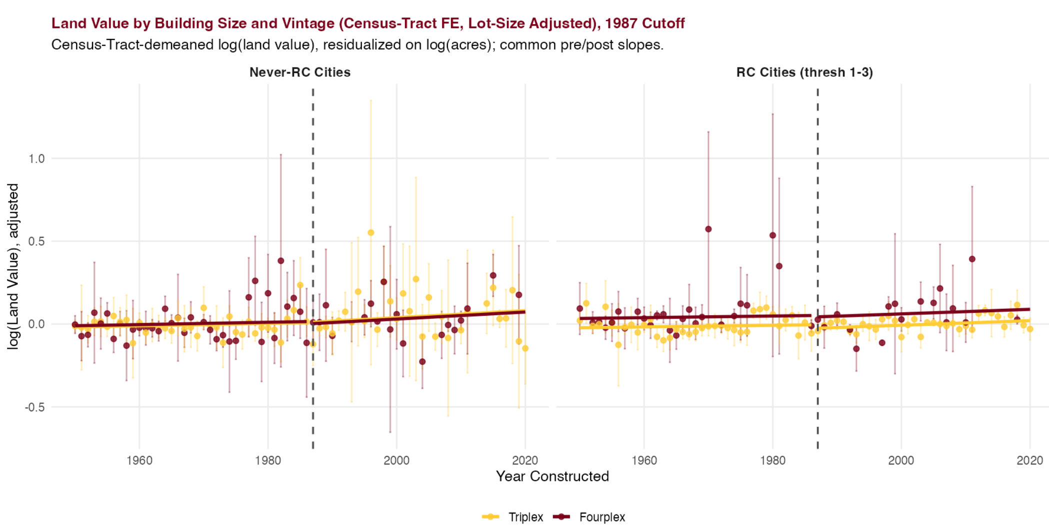
Table 11: Fourplex Treated Cell by Coverage Threshold

	N		Mean Land ($)		Mean Acres	
Thresh	Pre-87	Post-87	Pre-87	Post-87	Pre-87	Post-87
1	2,360	84	228,399	125,573	73.437	0.067
2	192	5	93,747	228,760	0.135	0.214
3	1,777	41	194,432	283,995	0.392	0.129
4	1,147	26	144,366	194,673	0.229	0.131
5	1,444	20	268,027	189,655	0.153	0.075

Notes: RC cities with validated thresholds; the DDD lead specification pools thresholds 1–3.

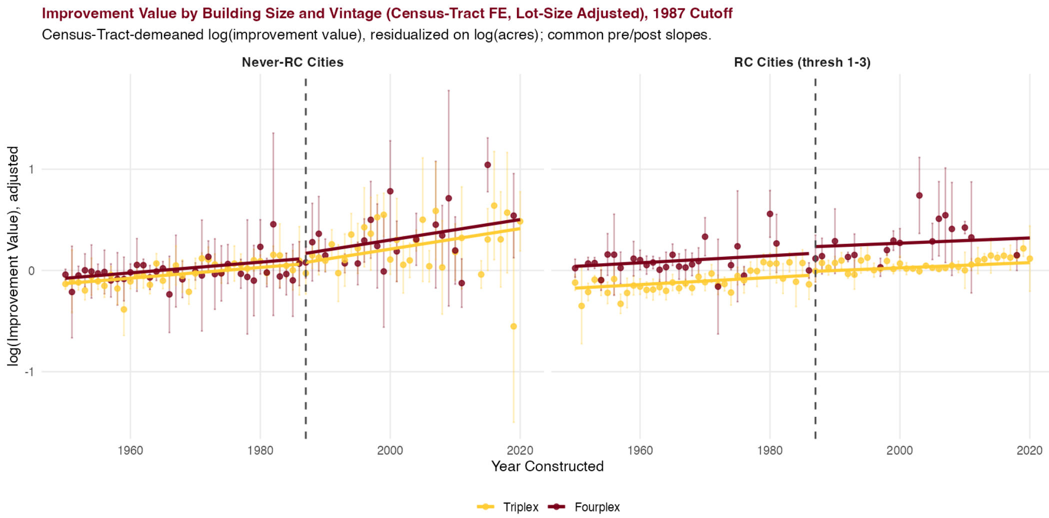
B Census-Tract Bin Scatters at Alternative Cutoffs (Lot-Size Adjusted)

Figure 3: Land Values by Building Size and Vintage: Lot-Size-Adjusted



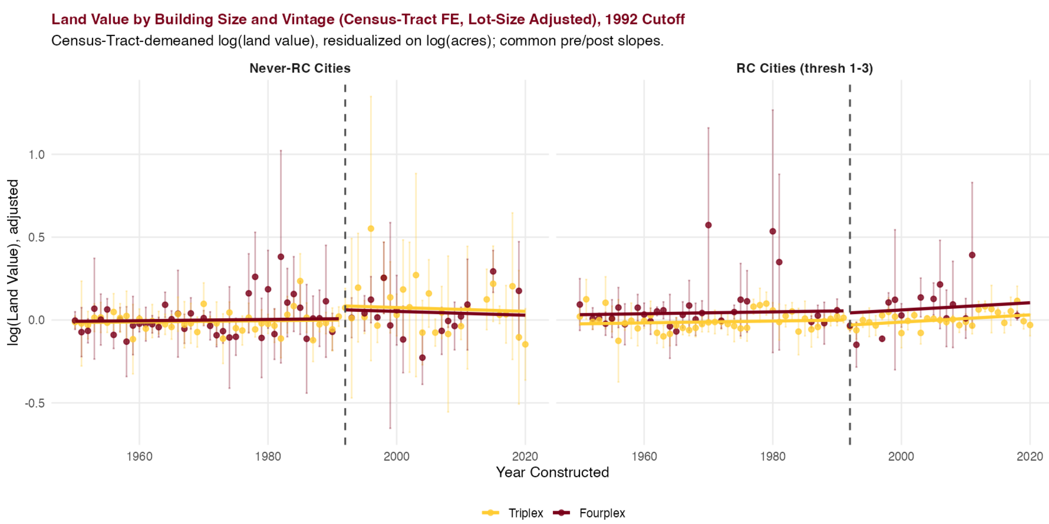
Notes: Census-tract-demeaned log(land value) residualized on log(acres) within tract, with common pre/post-1987 vintage slopes (matching Equation 2).

Figure 4: Improvement Values by Building Size and Vintage: Lot-Size-Adjusted (Census-Tract FE), 1987 Cutoff



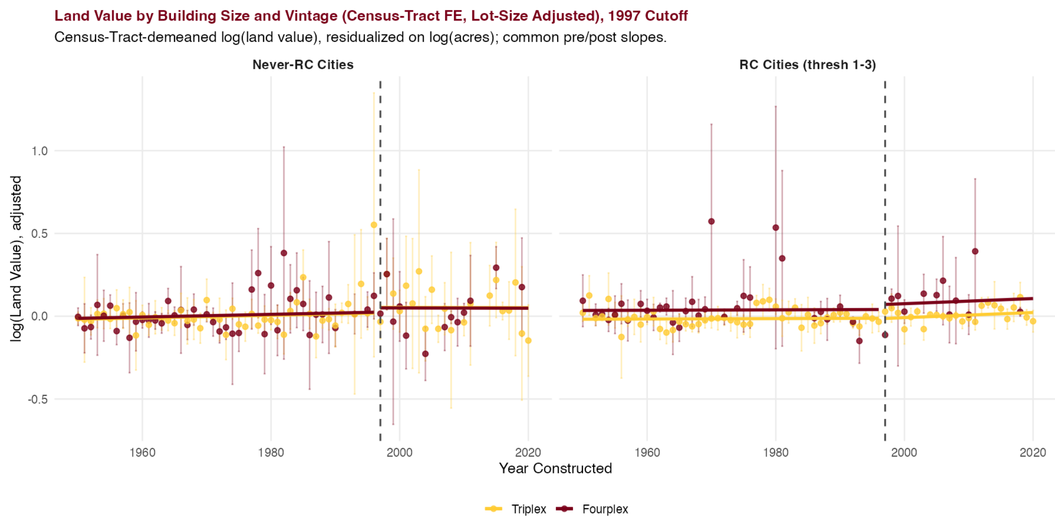
Notes: Census-tract-demeaned log(improvement value) residualized on log(acres), common pre/post slopes (as in Figure 3). Cutoffs 1992/1997 in Appendix B.

Figure 5: Land Values: Lot-Size-Adjusted (Census-Tract FE), 1992 Cutoff



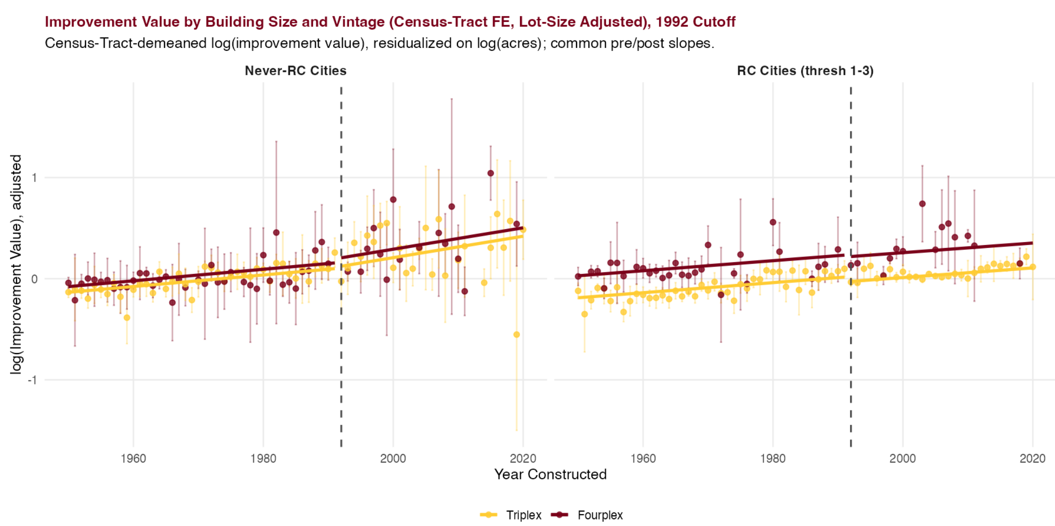
Notes: Census-tract-FE, lot-size-adjusted; same specification as Figure 3. Dashed line marks 1992.

Figure 6: Land Values: Lot-Size-Adjusted (Census-Tract FE), 1997 Cutoff



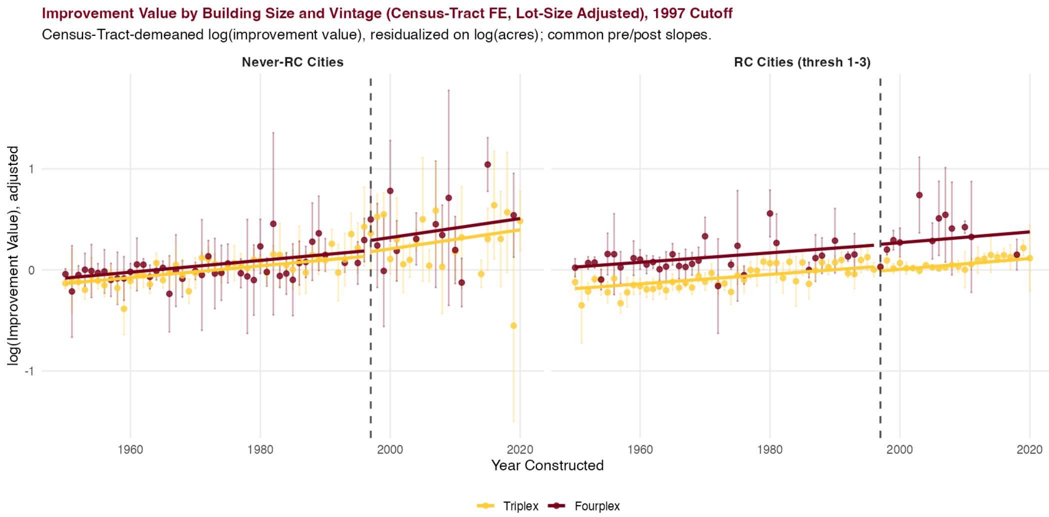
Notes: Census-tract-FE, lot-size-adjusted; same specification as Figure 3. Dashed line marks 1997.

Figure 7: Improvement Values: Lot-Size-Adjusted (Census-Tract FE), 1992 Cutoff



Notes: Census-tract-demeaned, lot-size-adjusted log(improvement value); same specification as Figure 4. Dashed line marks the 1992 cutoff.

Figure 8: Improvement Values: Lot-Size-Adjusted (Census-Tract FE), 1997 Cutoff



Notes: Census-tract-demeaned, lot-size-adjusted log(improvement value); same specification as Figure 4. Dashed line marks the 1997 cutoff.

C Full Census-Tract-FE Coefficient Estimates

Table 12: Full DDD Estimates: 1987 Cutoff (Census-Tract FE)

	Total	Improvement	Land
δ^{DDD}: Post $\times$ Four $\times$ RC	0.040	0.005	0.086
	(0.072)	(0.106)	(0.060)
Post $\times$ Fourplex	0.063	0.150*	−0.026
	(0.065)	(0.089)	(0.056)
Post $\times$ RC	−0.093**	−0.202***	−0.168***
	(0.043)	(0.062)	(0.040)
Fourplex $\times$ RC	0.042***	0.036**	−0.001
	(0.014)	(0.018)	(0.014)
Post	0.301***	0.502***	0.124***
	(0.043)	(0.063)	(0.041)
Fourplex	0.120***	0.166***	0.038***
	(0.011)	(0.015)	(0.011)
log(acres)	0.134***	0.108***	0.178***
	(0.024)	(0.019)	(0.032)
$\tilde{Y}^{-}$	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^{+}$	0.010***	0.010***	0.002***
	(0.001)	(0.002)	(0.001)
Diagnostics			
N	29,288	29,288	29,288
R^2	0.862	0.699	0.953
Within R^2	0.240	0.190	0.145
Census-tract FE	Yes	Yes	Yes

Notes: Census-tract FE version of Table 23. RC main effect absorbed by tract FE. Tract-assigned subset; standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 13: Full DDD Estimates: 1992 Cutoff (Census-Tract FE)

	Total	Improvement	Land
δ^{DDD}: Post × Four × RC	0.034	0.021	0.063
	(0.087)	(0.131)	(0.069)
Post × Fourplex	0.067	0.142	−0.012
	(0.081)	(0.115)	(0.066)
Post × RC	−0.112**	−0.246***	−0.189***
	(0.051)	(0.072)	(0.045)
Fourplex × RC	0.039***	0.030*	−0.001
	(0.014)	(0.018)	(0.014)
Post	0.358***	0.581***	0.161***
	(0.049)	(0.072)	(0.046)
Fourplex	0.122***	0.170***	0.038***
	(0.011)	(0.015)	(0.011)
log(acres)	0.135***	0.110***	0.179***
	(0.024)	(0.019)	(0.032)
$\tilde{Y}^-$	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^+$	0.010***	0.010***	0.002**
	(0.001)	(0.002)	(0.001)
Diagnostics			
N	29,288	29,288	29,288
R^2	0.861	0.696	0.953
Within R^2	0.234	0.182	0.145
Census-tract FE	Yes	Yes	Yes

Notes: Census-tract FE version of Table 24. RC main effect absorbed by tract FE. Tract-assigned subset; standard errors clustered at city level. $^*p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 14: Full DDD Estimates: 1997 Cutoff (Census-Tract FE)

	Total	Improvement	Land
δ^{DDD}: Post × Four × RC	0.026	0.028	0.057
	(0.093)	(0.143)	(0.074)
Post × Fourplex	0.095	0.167	0.015
	(0.087)	(0.125)	(0.071)
Post × RC	-0.113^{**}	-0.263^{***}	-0.174^{***}
	(0.057)	(0.080)	(0.046)
Fourplex × RC	0.038^{***}	0.029	-0.000
	(0.013)	(0.018)	(0.014)
Post	0.389^{***}	0.619^{***}	0.154^{***}
	(0.056)	(0.080)	(0.047)
Fourplex	0.121^{***}	0.169^{***}	0.037^{***}
	(0.011)	(0.015)	(0.011)
log(acres)	0.136^{***}	0.111^{***}	0.179^{***}
	(0.024)	(0.019)	(0.032)
$\tilde{Y}^{-}$	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^{+}$	0.010^{***}	0.011^{***}	0.002^{*}
	(0.002)	(0.002)	(0.001)
Diagnostics			
N	29,288	29,288	29,288
R^2	0.860	0.693	0.953
Within R^2	0.228	0.174	0.145
Census-tract FE	Yes	Yes	Yes

Notes: Census-tract FE version of Table 25. RC main effect absorbed by tract FE. Tract-assigned subset; standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 15: Full Within-RC DD Estimates: 1987 Cutoff (Census-Tract FE)

	Total	Improvement	Land
Post × Fourplex	0.090***	0.138**	0.046**
	(0.032)	(0.059)	(0.017)
Post	0.192***	0.294***	−0.017
	(0.029)	(0.053)	(0.017)
Fourplex	0.166***	0.205***	0.044***
	(0.008)	(0.009)	(0.010)
log(acres)	0.095***	0.076***	0.130***
	(0.020)	(0.016)	(0.036)
$\tilde{Y}^-$	0.001**	0.001**	0.000
	(0.000)	(0.001)	(0.000)
$\tilde{Y}^+$	0.008***	0.007***	0.002**
	(0.002)	(0.002)	(0.001)
Diagnostics			
N	19,556	19,556	19,556
R^2	0.852	0.707	0.961
Within R^2	0.249	0.223	0.092
Census-tract FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1987 cutoff, census-tract FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 16: Full Within-RC DD Estimates: 1992 Cutoff (Census-Tract FE)

	Total	Improvement	Land
Post × Fourplex	0.089^{**}	0.151^{**}	0.037^{**}
	(0.033)	(0.063)	(0.015)
Post	0.198^{***}	0.277^{***}	-0.016
	(0.032)	(0.060)	(0.016)
Fourplex	0.165^{***}	0.203^{***}	0.044^{***}
	(0.009)	(0.009)	(0.010)
log(acres)	0.095^{***}	0.076^{***}	0.130^{***}
	(0.020)	(0.016)	(0.036)
$\tilde{Y}^{-}$	0.001^{**}	0.001^{**}	0.000
	(0.000)	(0.001)	(0.000)
$\tilde{Y}^{+}$	0.009^{***}	0.008^{***}	0.002^{**}
	(0.002)	(0.003)	(0.001)
Diagnostics			
N	19,556	19,556	19,556
R^2	0.852	0.705	0.961
Within R^2	0.244	0.217	0.092
Census-tract FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1992 cutoff, census-tract FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 17: Full Within-RC DD Estimates: 1997 Cutoff (Census-Tract FE)

	Total	Improvement	Land
Post × Fourplex	0.112***	0.189**	0.057***
	(0.034)	(0.073)	(0.015)
Post	0.200***	0.246***	−0.005
	(0.034)	(0.064)	(0.013)
Fourplex	0.164***	0.201***	0.044***
	(0.008)	(0.009)	(0.010)
log(acres)	0.095***	0.077***	0.130***
	(0.020)	(0.016)	(0.036)
$\tilde{Y}^-$	0.001**	0.002**	0.000
	(0.000)	(0.001)	(0.000)
$\tilde{Y}^+$	0.010***	0.010***	0.002**
	(0.002)	(0.003)	(0.001)
Diagnostics			
N	19,556	19,556	19,556
R^2	0.851	0.702	0.961
Within R^2	0.240	0.211	0.092
Census-tract FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1997 cutoff, census-tract FE. Standard errors clustered at city level. $^*p < .1$, $^{**}p < .05$, $^{***}p < .01$.

D City Fixed-Effects Estimates

Table 18: DDD Results: Threshold 1–3 Cities (Lead Specification)

Cutoff	Outcome	DD (within-RC)	DD (never-RC)	$\hat{\delta}^{DDD}$
1987	log(Total)	0.104***	0.024	0.102
		(0.029)	(0.068)	(0.077)
	log(Imp)	0.124**	0.113	0.029
		(0.049)	(0.083)	(0.096)
	log(Land)	0.169*	−0.182*	0.401***
		(0.087)	(0.105)	(0.148)
1992	log(Total)	0.102***	0.009	0.113
		(0.030)	(0.081)	(0.088)
	log(Imp)	0.143***	0.092	0.063
		(0.049)	(0.105)	(0.116)
	log(Land)	0.150*	−0.208	0.408**
		(0.089)	(0.140)	(0.171)
1997	log(Total)	0.124***	0.012	0.130
		(0.037)	(0.089)	(0.099)
	log(Imp)	0.176***	0.099	0.087
		(0.056)	(0.120)	(0.132)
	log(Land)	0.176	−0.204	0.424**
		(0.115)	(0.154)	(0.201)
Diagnostics				
N		19,770	30,351	
RC cities		43	43	
Never-RC cities		—	376	
Fixed Effects		City (43)	City (416)	
Clustering		City	City	
Within R^2		0.257 / 0.229 / 0.032	0.257 / 0.201 / 0.068	
Dep. var. mean		12.57 / 12.29 / 10.93	12.66 / 12.29 / 11.20	

Notes: DD (within-RC) is the Post × Fourplex coefficient from a regression on RC cities only (no never-RC baseline). DD (never-RC) and $\hat{\delta}^{DDD}$ are from Equation 2 pooling RC and never-RC cities: DD (never-RC) is Post × Fourplex at RC= 0; $\hat{\delta}^{DDD}$ is the triple interaction. City FE. Within R^2 and dep. var. mean as Total / Imp / Land. Clustered at city. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 19: DDD Robustness: $\hat{\delta}^{DDD}$ on log(Land) Across Threshold Groups

Sample / Specification	1987	1992	1997
Thresh 1–3 pooled (lead, 43 cities)	0.401***	0.408**	0.424**
	(0.148)	(0.171)	(0.201)
Thresh=3 only (21 cities)	0.281**	0.323**	0.313*
	(0.111)	(0.151)	(0.168)
Thresh=1 only (17 cities)	0.467***	0.422**	0.482**
	(0.165)	(0.192)	(0.234)
Thresh 1–2 (22 cities)	0.470***	0.421**	0.471**
	(0.158)	(0.186)	(0.227)
Diagnostics			
Fixed Effects		City	
Clustering		City	

Notes: Each row reports $\hat{\delta}^{DDD}$ from Equation 2 on log(land), pooling RC and never-RC cities. City FE; clustered at city. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 20: Leave-One-Out DDD Robustness: $\hat{\delta}^{DDD}$ on log(Land), Threshold 1–3

	1987	1992	1997
Baseline $\hat{\delta}^{DDD}$	0.401***	0.408**	0.424**
14 treated cities (have post-87 fourplexes):			
Sig at 5%	13/14	13/14	13/14
Range	[0.222, 0.468]	[0.223, 0.487]	[0.201, 0.514]
29 counterfactual-only cities:			
Sig at 5%	29/29	29/29	29/29
Range	[0.388, 0.415]	[0.394, 0.422]	[0.410, 0.441]
Jackknife t-stat	2.05	2.02	1.76

Notes: Each row drops one RC city at a time and re-estimates the DDD from Equation 2 on log(land value) at the threshold 1–3 pooled lead specification (43 RC cities + 376 never-RC). Treated cities contain post-1987 fourplexes that identify δ^{DDD}; counterfactual-only cities contribute to the lower-order interactions. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 21: DDD Robustness: Alternative Treated Group Definitions

Treated group	1987	1992	1997
Fourplexes only (Class 2)	0.401***	0.408**	0.424**
4–5 units (Class 2 + 4C)	0.247	0.210	0.187
4+ units (all, incl. 4C)	0.332*	0.347*	0.304
Diagnostics			
Fixed Effects		City	
Clustering		City	

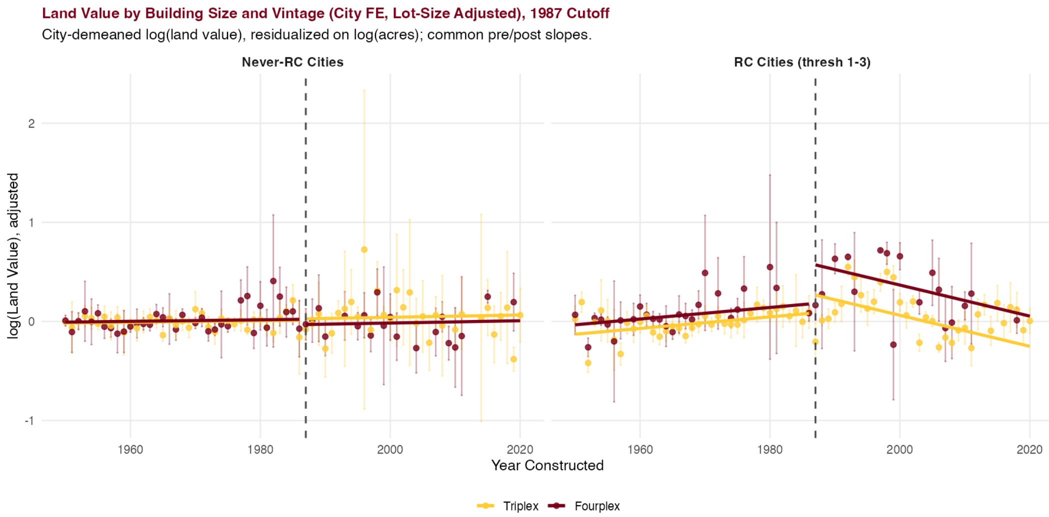
Notes: Each cell reports $\hat{\delta}^{DDD}$ from Equation 2 on log(land) at the 1987, 1992, and 1997 cutoffs, threshold 1–3 sample; control group triplexes. City FE; clustered at city. Post-cutoff treated cells (1987/1992/1997): 127/109/90 (fourplexes), 228/203/179 (4–5 units), 427/393/362 (4+ units). $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 22: Value Placebo: $\hat{\delta}^{DDD}$ on log(Land) at Pre-NCE Cutoffs

Placebo cutoff	Treated cell	$\hat{\delta}^{DDD}$	SE	p
1965	99	+0.001	0.056	0.989
1970	49	+0.042	0.068	0.532
1975	32	−0.041	0.100	0.681
1980	19	−0.037	0.150	0.808
1987 (true)	130	+0.401***	0.148	0.007

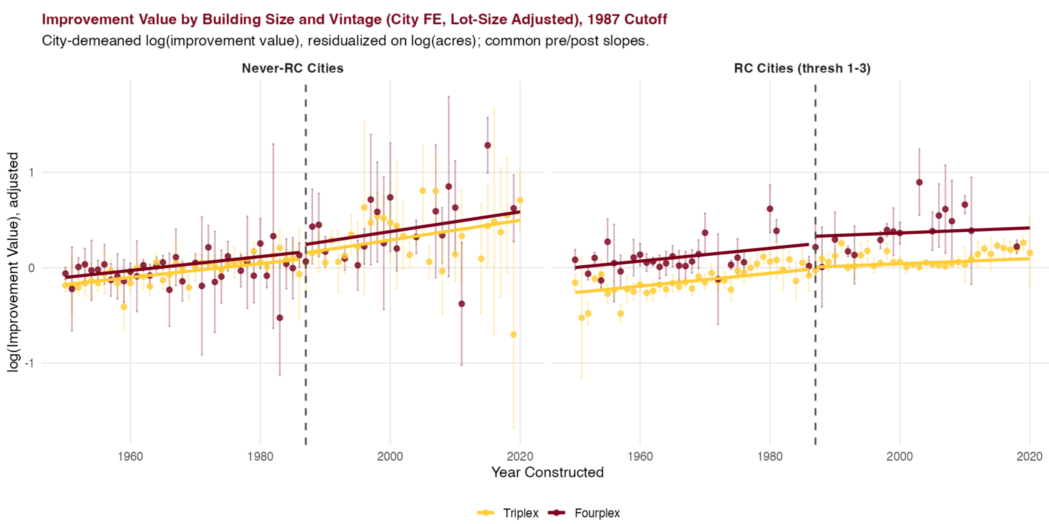
Notes: Each row re-estimates Equation 2 on log(land value) at a placebo cutoff, restricting to pre-1987 buildings (none NCE-eligible); $N = 27{,}426$, city FE, clustered at city. The final row is the true 1987 estimate. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Figure 9: Land Values by Building Size and Vintage: Lot-Size-Adjusted (City FE)



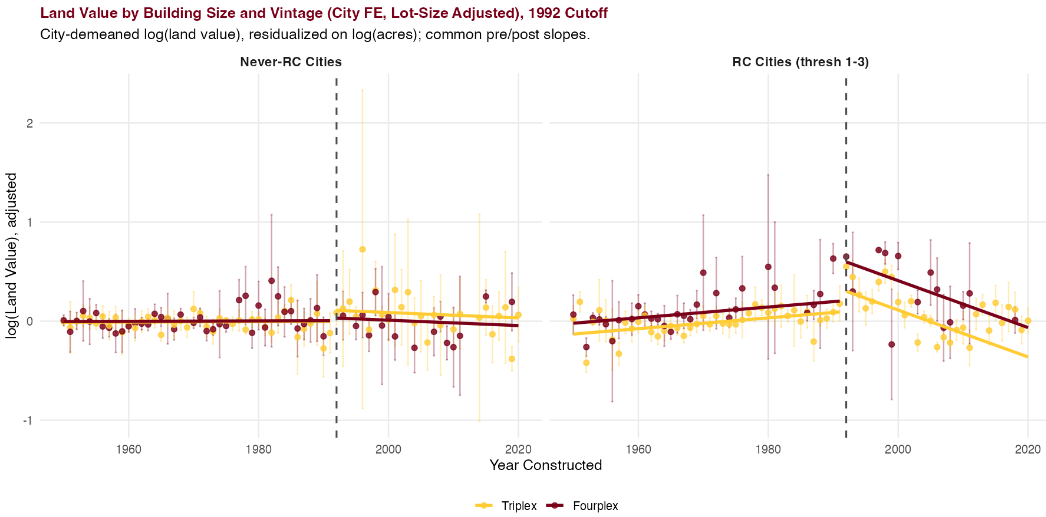
Notes: City-demeaned log(land value) residualized on log(acres) within city, with common pre/post-1987 vintage slopes. RC cities (right) and never-RC (left).

Figure 10: Improvement Values by Building Size and Vintage: Lot-Size-Adjusted (City FE), 1987 Cutoff



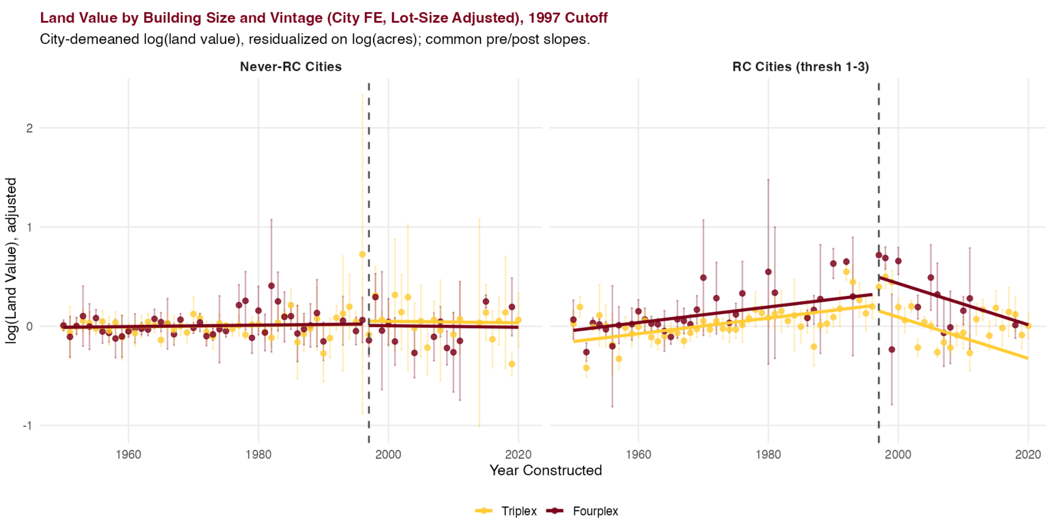
Notes: City-demeaned log(improvement value) residualized on log(acres), common pre/post slopes (as in Figure 9).

Figure 11: Land Values: Lot-Size-Adjusted (City FE), 1992 Cutoff



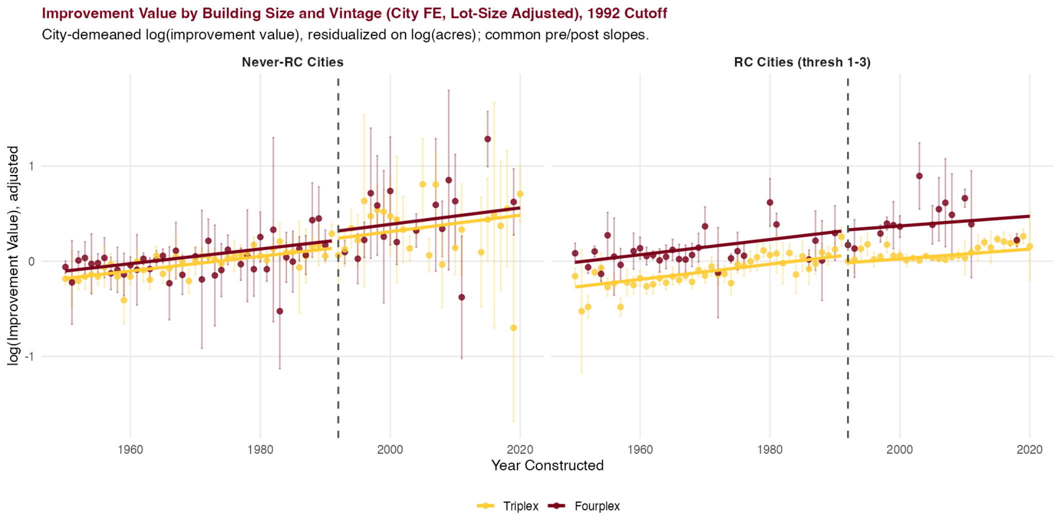
Notes: City-demeaned, lot-size-adjusted; same specification as Figure 9, 1992 cutoff.

Figure 12: Land Values: Lot-Size-Adjusted (City FE), 1997 Cutoff



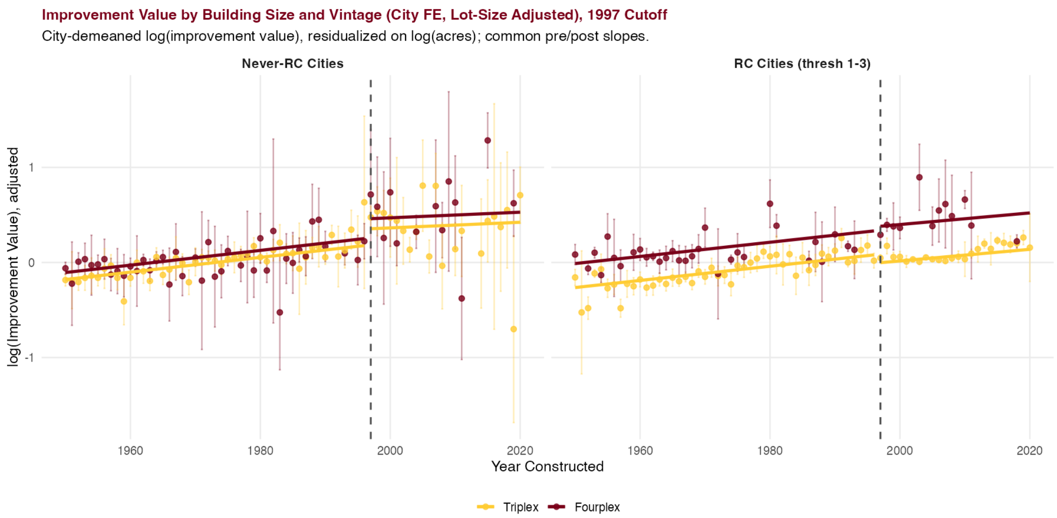
Notes: City-demeaned, lot-size-adjusted; same specification as Figure 9, 1997 cutoff.

Figure 13: Improvement Values: Lot-Size-Adjusted (City FE), 1992 Cutoff



Notes: City-demeaned, lot-size-adjusted; same specification as Figure 10, 1992 cutoff.

Figure 14: Improvement Values: Lot-Size-Adjusted (City FE), 1997 Cutoff



Notes: City-demeaned, lot-size-adjusted; same specification as Figure 10, 1997 cutoff.

Table 23: Full DDD Estimates: 1987 Cutoff

	Total	Improvement	Land
δ^{DDD}: Post × Four × RC	0.102	0.029	0.401***
	(0.077)	(0.096)	(0.148)
Post × Fourplex	0.024	0.113	−0.182*
	(0.068)	(0.083)	(0.105)
Post × RC	−0.118**	−0.248***	−0.049
	(0.049)	(0.055)	(0.074)
Fourplex × RC	0.052***	0.038**	0.052*
	(0.015)	(0.019)	(0.031)
Post	0.369***	0.528***	0.327**
	(0.047)	(0.056)	(0.139)
Fourplex	0.116***	0.162***	0.034***
	(0.011)	(0.015)	(0.012)
log(acres)	0.156***	0.135***	0.166***
	(0.027)	(0.020)	(0.055)
$\tilde{Y}^-$	0.000	0.000	−0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^+$	0.008***	0.010***	−0.008
	(0.001)	(0.002)	(0.007)
Diagnostics			
N	30,351	30,351	30,351
R^2	0.838	0.675	0.860
Within R^2	0.257	0.201	0.068
City FE	Yes	Yes	Yes

Notes: All coefficients from Equation 2 at the 1987 cutoff. RC main effect absorbed by city FE. Standard errors clustered at city level. $^*p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 24: Full DDD Estimates: 1992 Cutoff

	Total	Improvement	Land
δ^{DDD}: Post × Four × RC	0.113	0.063	0.408**
	(0.088)	(0.116)	(0.171)
Post × Fourplex	0.009	0.092	−0.208
	(0.081)	(0.105)	(0.140)
Post × RC	−0.155***	−0.301***	−0.111
	(0.057)	(0.063)	(0.074)
Fourplex × RC	0.049***	0.032*	0.053*
	(0.015)	(0.019)	(0.031)
Post	0.459***	0.619***	0.460***
	(0.057)	(0.063)	(0.143)
Fourplex	0.118***	0.166***	0.034***
	(0.011)	(0.015)	(0.012)
log(acres)	0.157***	0.136***	0.165***
	(0.027)	(0.020)	(0.055)
$\tilde{Y}^{-}$	0.000	0.000	−0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^{+}$	0.007***	0.010***	−0.015**
	(0.001)	(0.002)	(0.007)
Diagnostics			
N	30,351	30,351	30,351
R^2	0.837	0.672	0.861
Within R^2	0.251	0.194	0.070
City FE	Yes	Yes	Yes

Notes: All coefficients from Equation 2 at the 1992 cutoff. RC main effect absorbed by city FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 25: Full DDD Estimates: 1997 Cutoff

	Total	Improvement	Land
δ^{DDD}: Post × Four × RC	0.130	0.087	0.424**
	(0.099)	(0.132)	(0.201)
Post × Fourplex	0.012	0.099	−0.204
	(0.089)	(0.120)	(0.154)
Post × RC	−0.169***	−0.317***	−0.162**
	(0.057)	(0.070)	(0.064)
Fourplex × RC	0.047***	0.030	0.053*
	(0.015)	(0.019)	(0.031)
Post	0.485***	0.664***	0.403***
	(0.054)	(0.069)	(0.109)
Fourplex	0.117***	0.166***	0.032**
	(0.011)	(0.015)	(0.013)
log(acres)	0.157***	0.137***	0.165***
	(0.027)	(0.021)	(0.055)
$\tilde{Y}^{-}$	0.000	0.000	−0.000
	(0.000)	(0.000)	(0.000)
$\tilde{Y}^{+}$	0.007***	0.011***	−0.014**
	(0.001)	(0.002)	(0.006)
Diagnostics			
N	30,351	30,351	30,351
R^2	0.835	0.669	0.860
Within R^2	0.242	0.186	0.065
City FE	Yes	Yes	Yes

Notes: All coefficients from Equation 2 at the 1997 cutoff. RC main effect absorbed by city FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 26: Full Within-RC DD Estimates: 1987 Cutoff

	Total	Improvement	Land
Post × Fourplex	0.104***	0.124**	0.169*
	(0.029)	(0.049)	(0.087)
Post	0.241***	0.245***	0.422
	(0.039)	(0.057)	(0.259)
Fourplex	0.175***	0.202***	0.107***
	(0.011)	(0.011)	(0.038)
log(acres)	0.098***	0.092***	0.065
	(0.021)	(0.017)	(0.062)
$\tilde{Y}^-$	0.001**	0.001**	−0.001
	(0.000)	(0.001)	(0.001)
$\tilde{Y}^+$	0.006***	0.007***	−0.011
	(0.001)	(0.002)	(0.007)
Diagnostics			
N	19,769	19,769	19,769
R^2	0.828	0.682	0.825
Within R^2	0.257	0.229	0.032
City FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1987 cutoff, city FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 27: Full Within-RC DD Estimates: 1992 Cutoff

	Total	Improvement	Land
Post × Fourplex	0.102***	0.143***	0.150*
	(0.030)	(0.049)	(0.089)
Post	0.259***	0.226***	0.482**
	(0.038)	(0.053)	(0.198)
Fourplex	0.173***	0.200***	0.107***
	(0.012)	(0.011)	(0.038)
log(acres)	0.098***	0.091***	0.066
	(0.021)	(0.017)	(0.062)
$\tilde{Y}^{-}$	0.001***	0.002**	−0.001
	(0.000)	(0.001)	(0.001)
$\tilde{Y}^{+}$	0.006***	0.009***	−0.019***
	(0.001)	(0.002)	(0.005)
Diagnostics			
N	19,769	19,769	19,769
R^2	0.827	0.680	0.825
Within R^2	0.252	0.224	0.035
City FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1992 cutoff, city FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.

Table 28: Full Within-RC DD Estimates: 1997 Cutoff

	Total	Improvement	Land
Post × Fourplex	0.124***	0.176***	0.176
	(0.037)	(0.056)	(0.115)
Post	0.233***	0.214***	0.309**
	(0.030)	(0.057)	(0.118)
Fourplex	0.171***	0.198***	0.104***
	(0.011)	(0.010)	(0.036)
log(acres)	0.097***	0.090***	0.064
	(0.022)	(0.016)	(0.065)
$\tilde{Y}^-$	0.001***	0.002***	−0.001
	(0.000)	(0.001)	(0.001)
$\tilde{Y}^+$	0.007***	0.011***	−0.016***
	(0.001)	(0.002)	(0.004)
Diagnostics			
N	19,769	19,769	19,769
R^2	0.825	0.679	0.823
Within R^2	0.245	0.220	0.025
City FE	Yes	Yes	Yes

Notes: Within-RC DD (Post × Fourplex on RC cities only) at the 1997 cutoff, city FE. Standard errors clustered at city level. $^{*}p < .1$, $^{**}p < .05$, $^{***}p < .01$.